

Summer school on Machine Learning for Space

KTH Royal Institute of Technology

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**Università
di Genova**



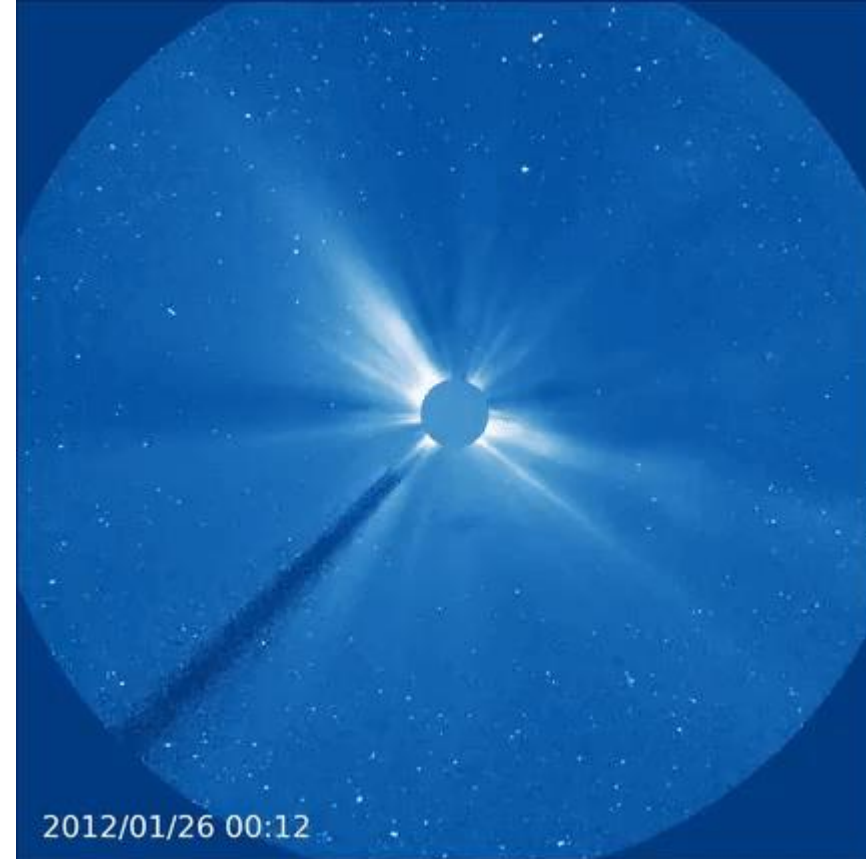
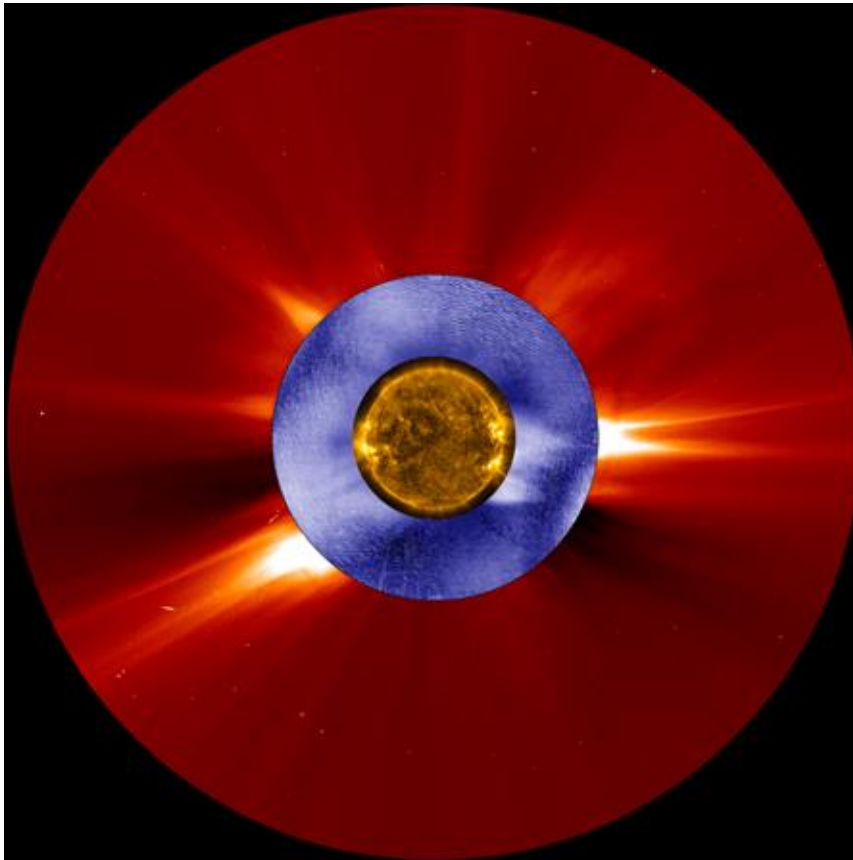
ASAP
Automatics in Space Exploration



Which are the main drivers of geomagnetic storms?

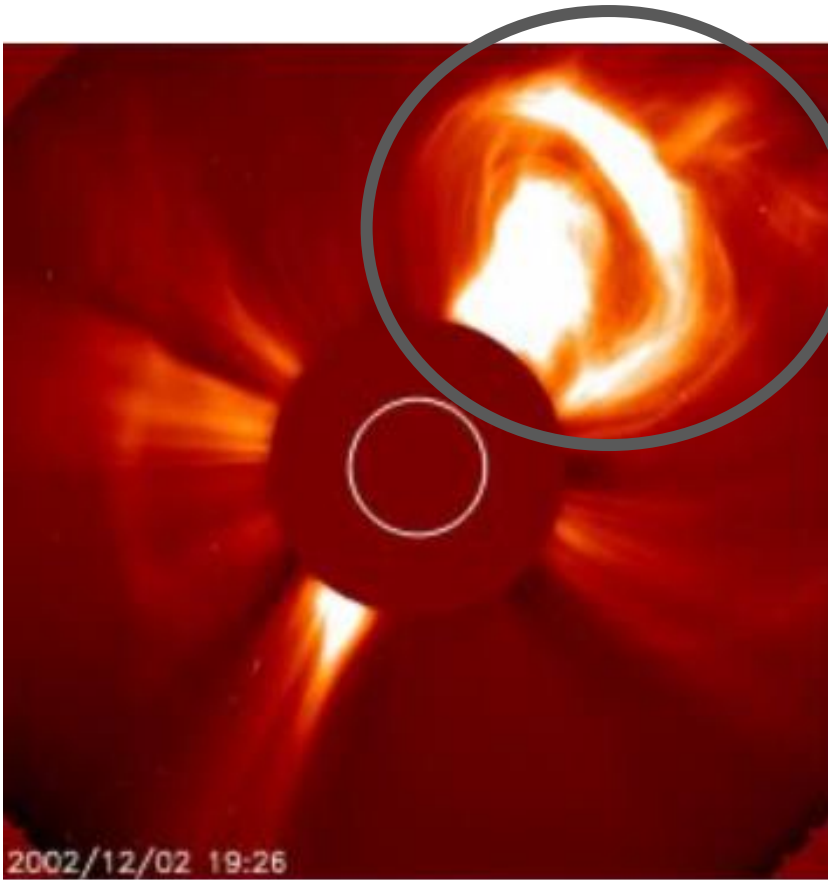
Coronal Mass Ejections (CMEs) are the main geo-effective events

Coronal Mass Ejections (CMEs) consist of large burst of plasma and magnetic field that erupts from the Sun and travels through space, and they are typically triggered by solar flares. These events can strongly impact satellites, power grids, and communications, so accurate prediction is crucial for the mitigation of these effects.



How can we predict travel time?

The observations of CMEs are typically performed by means of **remote-sensing instruments**



once a CME is detected, we rely on instruments like LASCO to extract parameters like the CME's initial propagation speed, its mass and initial cross section.



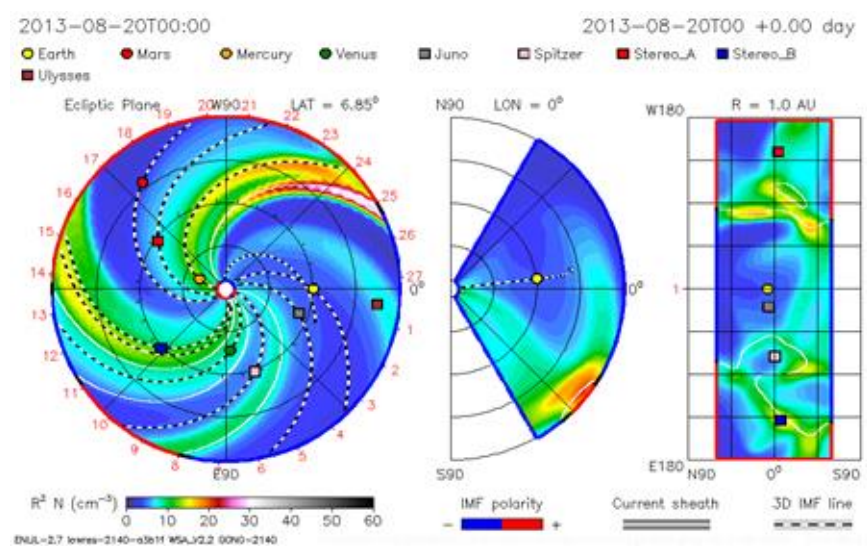
When will the CME reach the Earth?

Large Angle and Spectrometric Coronagraph (**LASCO**)
on board the Solar and Heliospheric Observatory (SOHO)

Predicting CME arrival: Physics-based or data-driven?

Physics-based approaches

Drag-based models, MHD-based equations ...

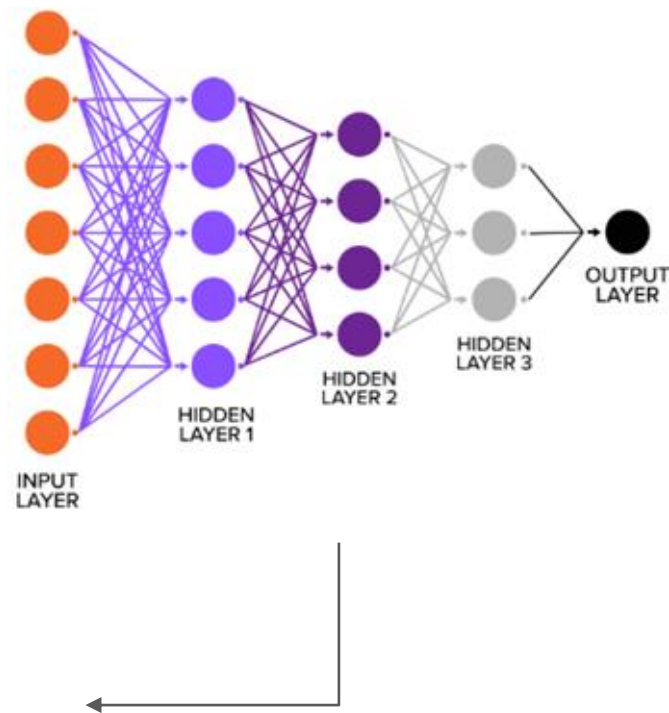


Typical mean absolute error in the prediction
~ 10-12 h

Physics-driven ML

Data-driven approaches

Machine/deep learning



Physics-Driven ML

We start from the **Drag Based Model**, which describes the evolution of a CME as a function of the solar wind conditions

$$\ddot{r}(t) = -\gamma|\dot{r}(t) - w|(\dot{r}(t) - w)$$

Drag parameter $\gamma = C \frac{A\rho}{m}$

ρ = solar wind density
 A = CME impact area
 m = CME mass
 C = drag coefficient

$\ddot{r}$ = CME acceleration
 $\dot{r}$ = CME speed
 w = solar wind speed

The Drag coefficient C is unknown $\rightarrow$ we use a NN to estimate C

Physics-Driven ML

We include the drag-based model in the loss function

λ is a regularization parameter

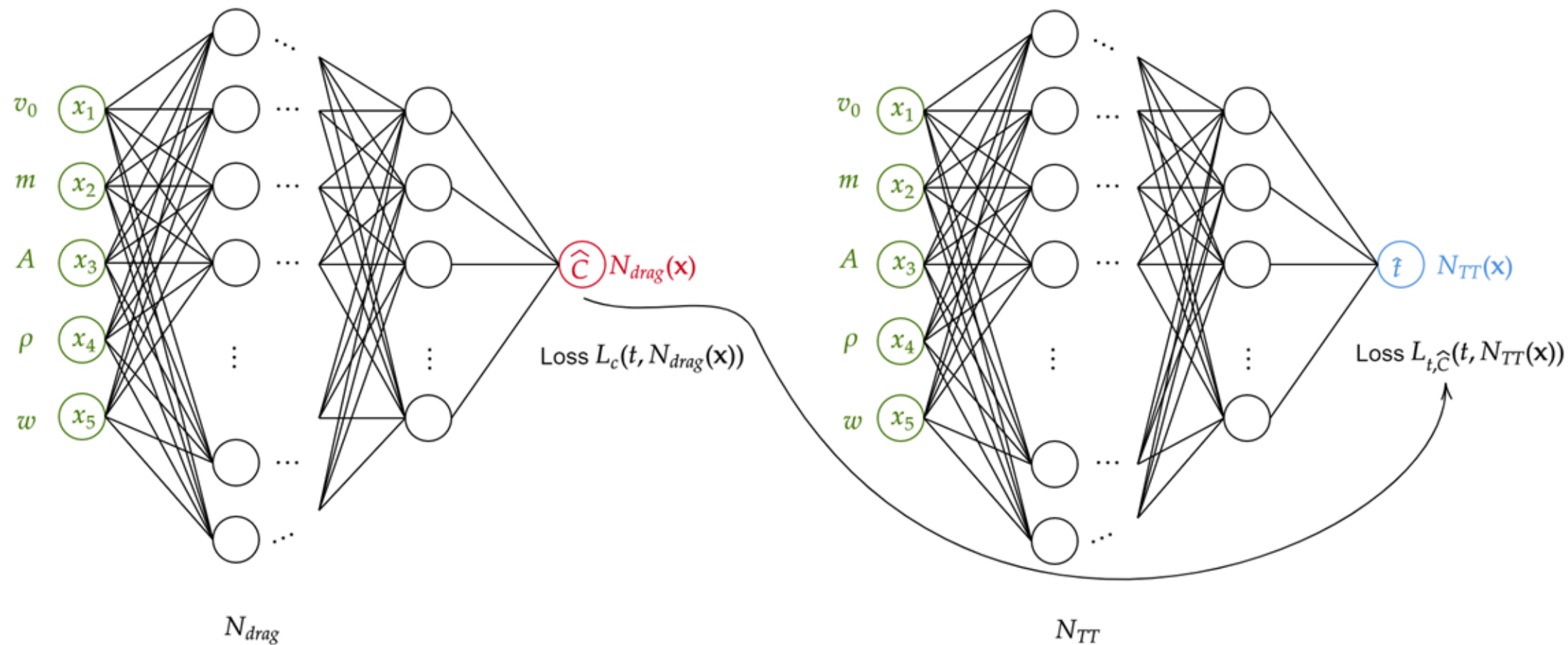
$$L_t(t, f(w, x)) = \lambda \boxed{(t - f(w, x))^2} + (1 - \lambda) \boxed{(1 - r(f(w, x), C))^2}$$

Data-driven term Physics-driven term
(MSE)

The loss than has **two** different **components**

- **standard data-driven term**, which minimizes the error between the predicted and the true travel time.
- **physics driven term**: enforces that the predicted travel time is consistent with the drag-based equation

Operational-Oriented Neural Networks



Remark

In the inference step we do not need the drag-parameter as input
→ the network can be used for real-time forecasting.

Dataset

- 123 CME events occurred in the time range between 1997 and 2018 [1]

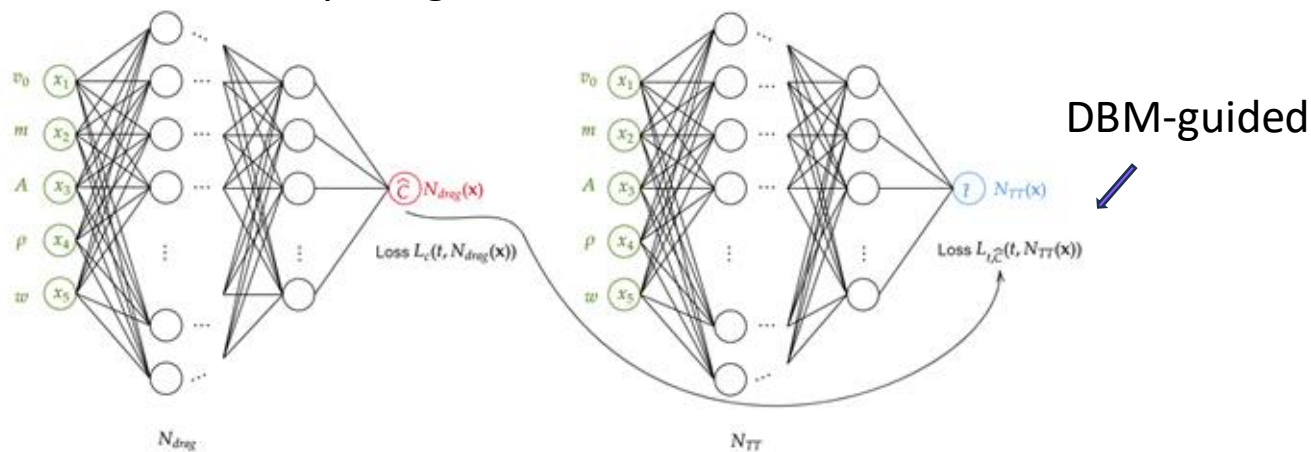
Name	Notation	Unity	Description	Source
CME height of eruption	r_0	km	$r_0 = 20 R_{\odot}$, $R_{\odot} = 6.957 \cdot 10^5$ km	-
CME time of eruption	t_0	s	eruption time on the Sun at r_0	(Napoletano et al. 2022)
CME Time of Arrival	ToA	s	estimated arrival time at 1 AU	R & C
CME Travel time	TT	s	estimated time between t_0 and ToA	R & C, (Napoletano et al. 2022)
CME initial speed	v_0	km/s	initial propagation speed from eruption	LASCO
CME mass	m	g	estimated CME mass	LASCO
CME impact area	A	km ²	CME impact area, constant angular width	LASCO
Solar wind density	ρ	g/km ³	mean over one hour after t_0	CELIAS
Solar wind speed	w	km/s	mean over one hour after t_0	CELIAS
Drag parameter	C	dimensionless	parameter of the drag based model	this work

- In order to perform a statistical assessment of the physics-driven machine learning approach to travel time prediction, we realized 100 random realizations of the training, validation, and test

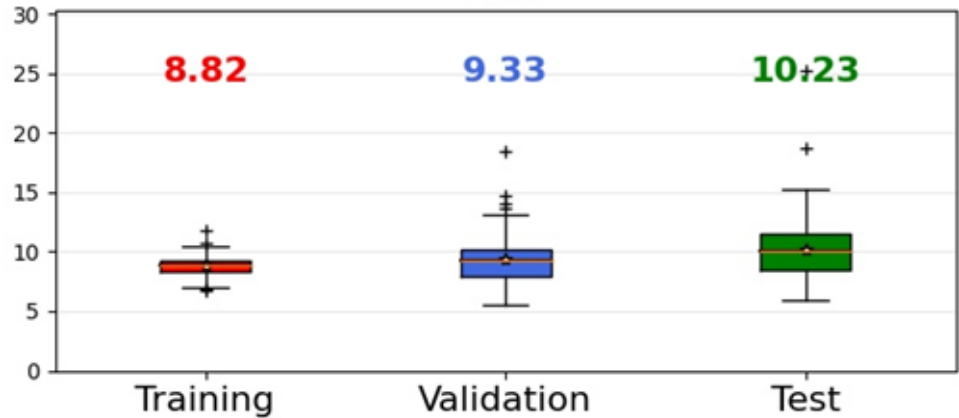
[1] Napoletano, G, et al. "Parameter Distributions for the Drag-Based Modeling of CME Propagation." *Space Weather* 20.9 (2022): e2021SW002925

Operational-Oriented Neural Networks

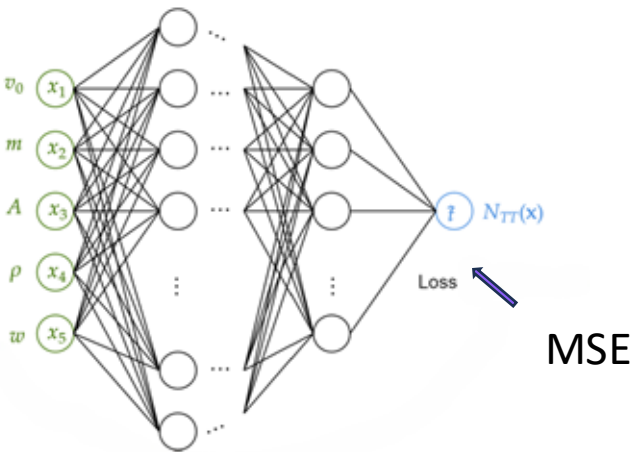
Physics-guided ML



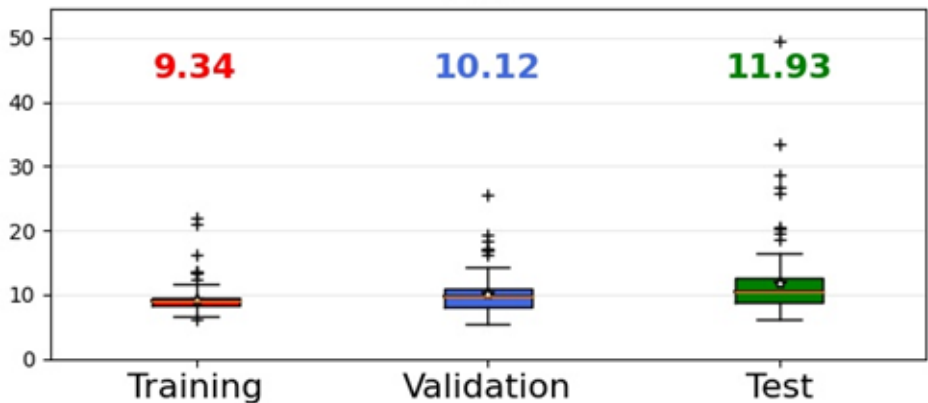
MAE



Fully data-driven ML

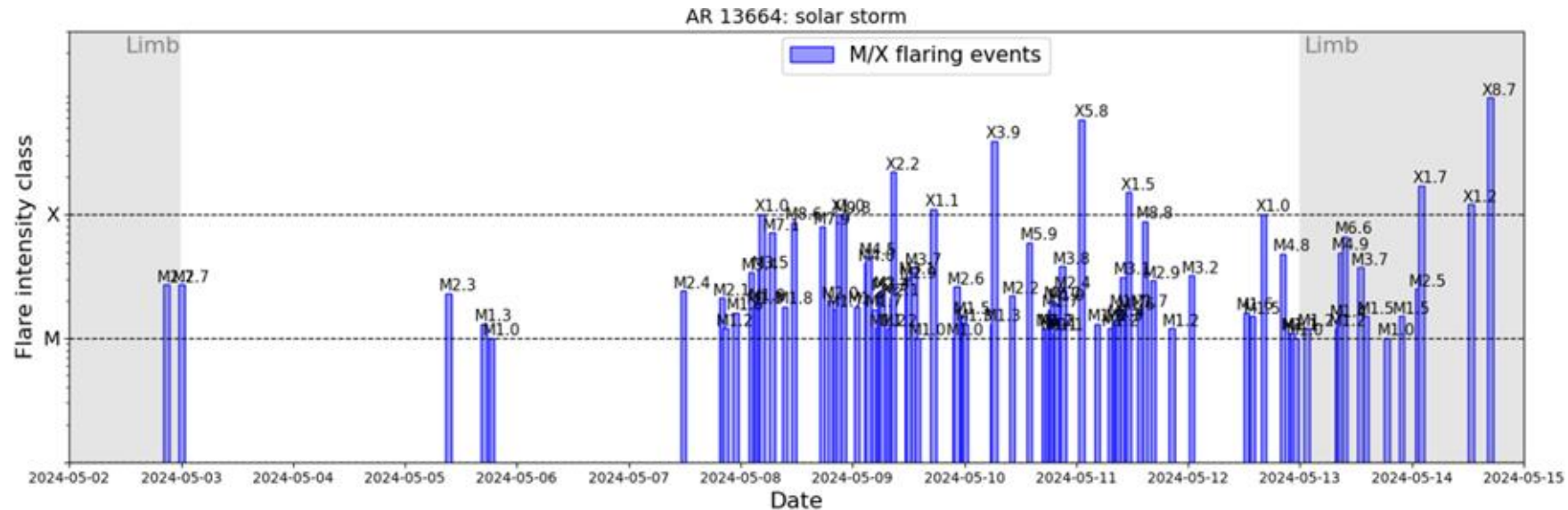
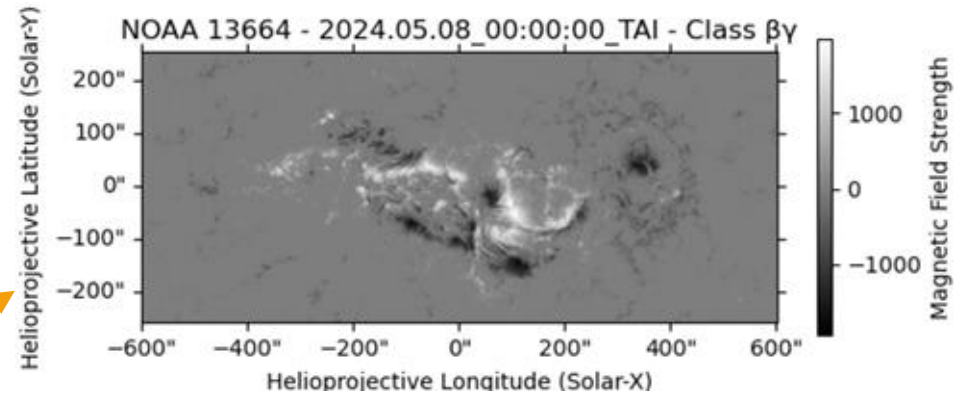
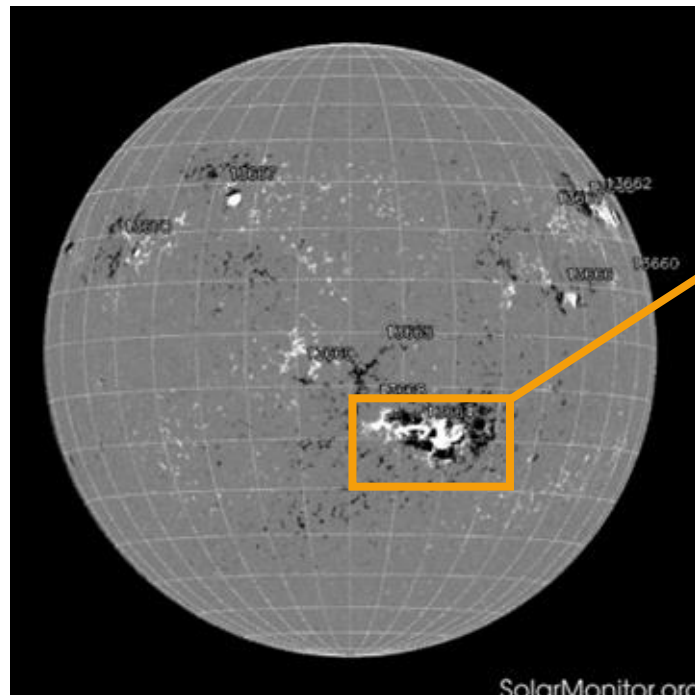


MAE

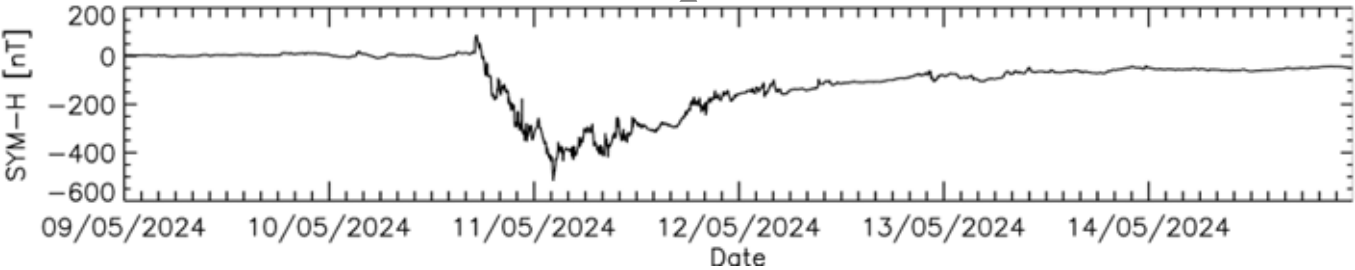
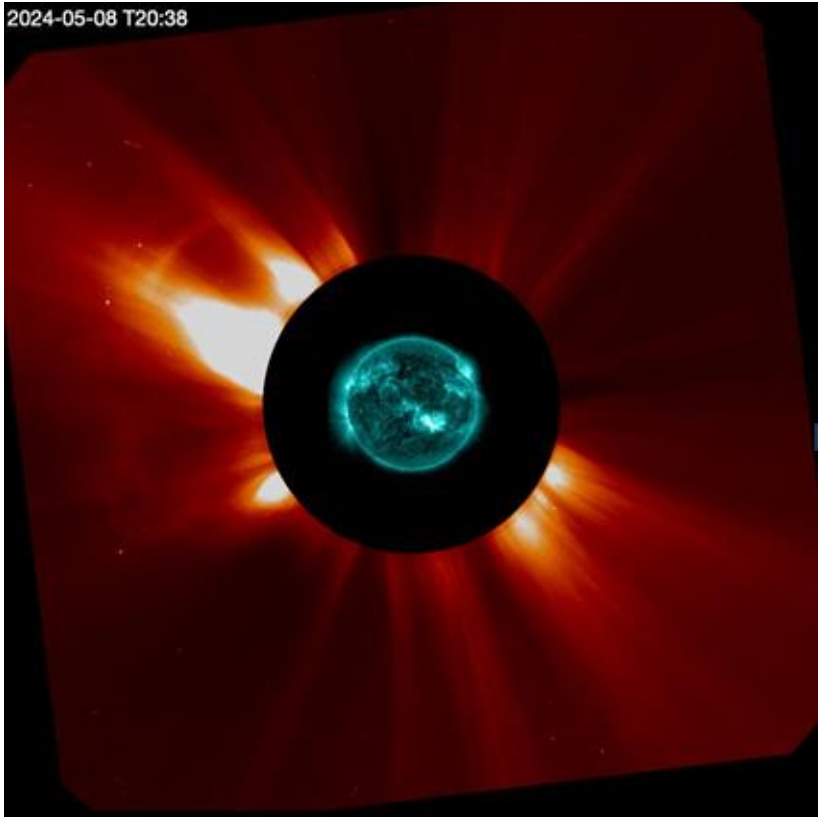


May 2024 Superstorm

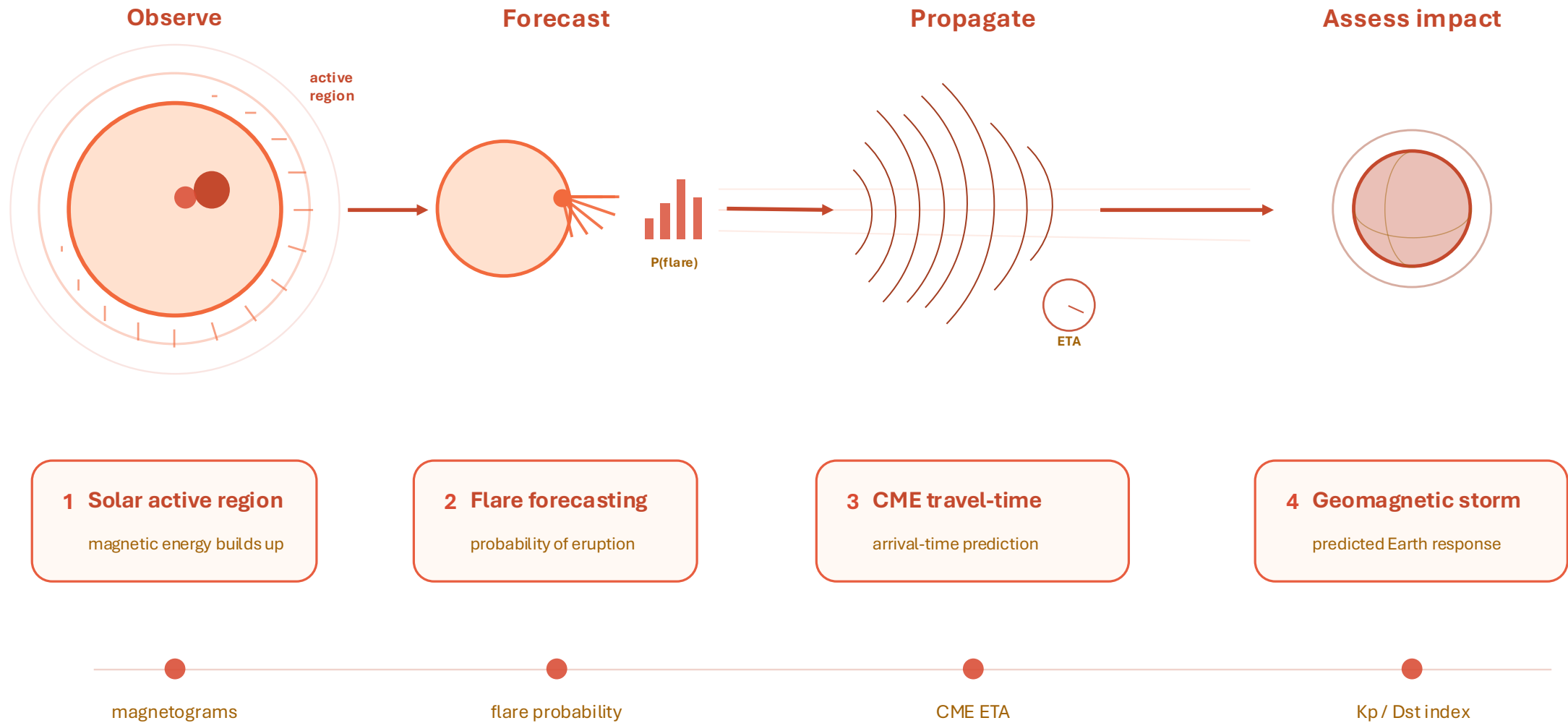
In 2024 May 10/11, the second strongest geomagnetic storm has occurred in the space era, with a peak Dst index of lower than -400 nT. The storm was caused by NOAA Active Region (AR) 13664, which was the source of a large number of coronal mass ejections and flares, including more than 10 X-class flares.



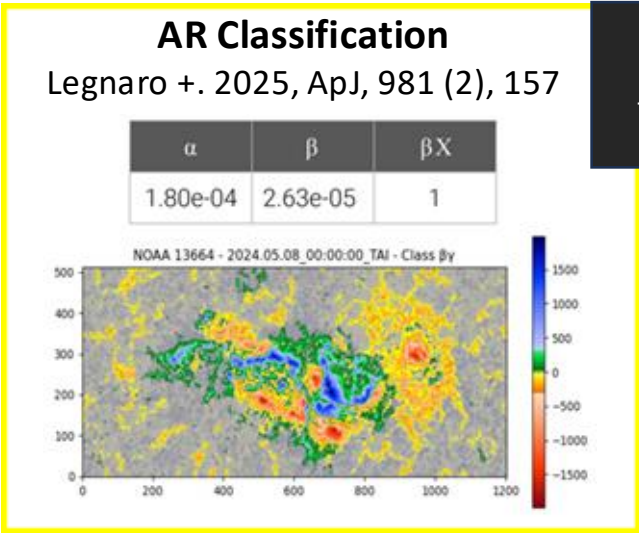
May 2024 Superstorm



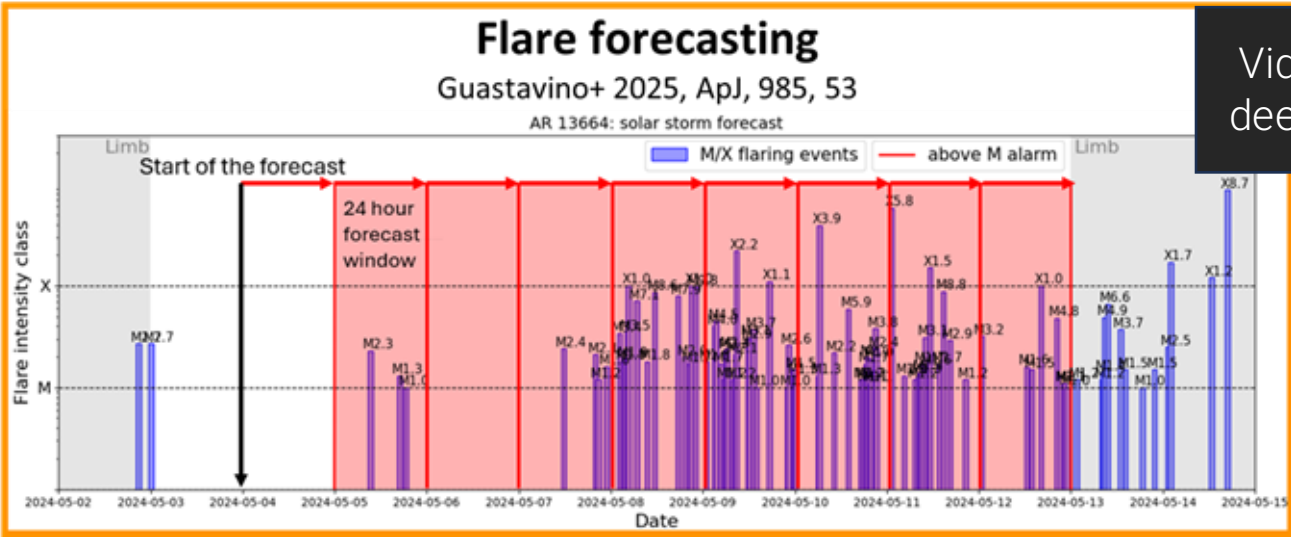
Sun-to-Earth Space Weather Prediction Chain



May 2024 Superstorm



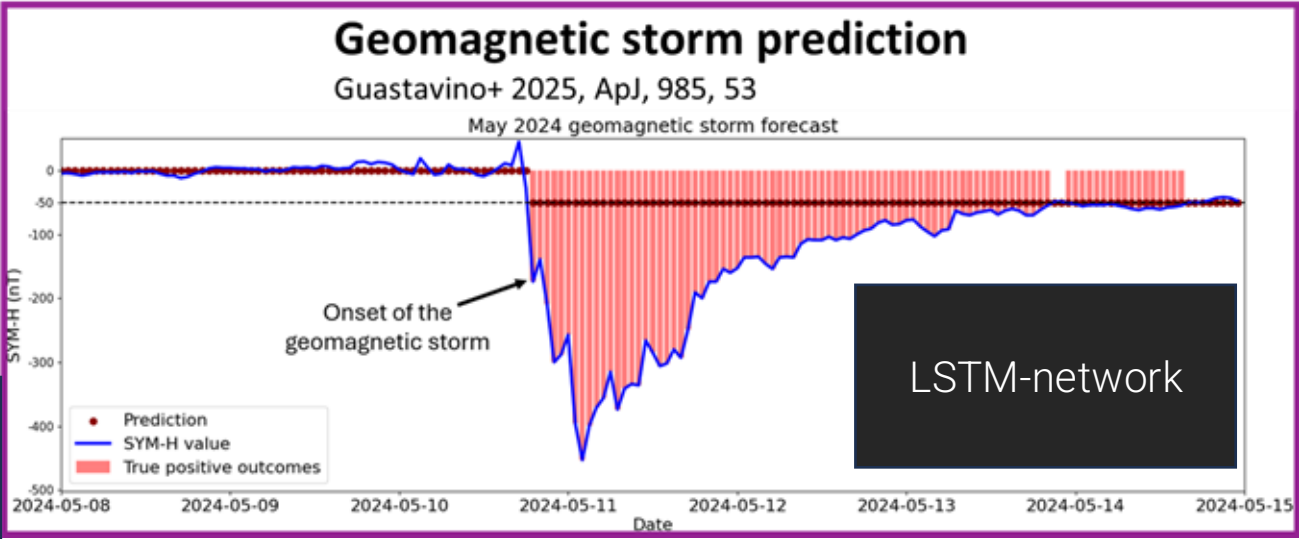
Vision transformer



Video-based deep learning



Physics-driven neural network



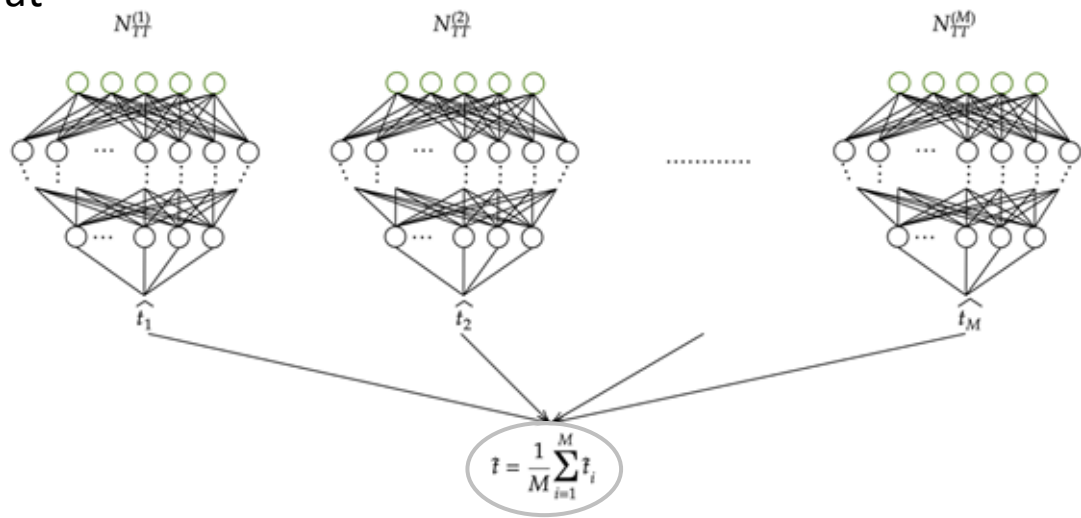
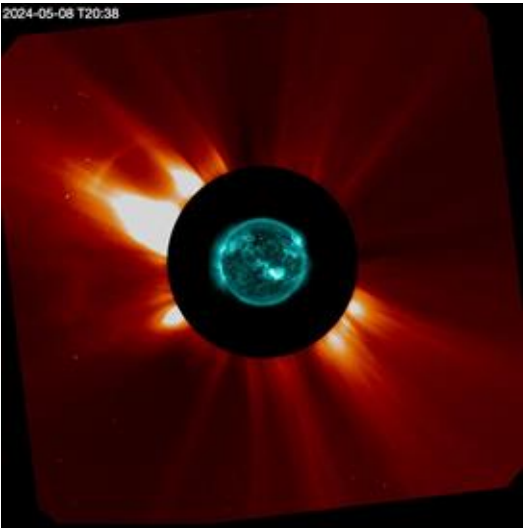
LSTM-network

CME's travel time prediction

Input parameter extraction at
t0 = 2024-05-08 T22:01

v0 = 2598.65 km/s
m = 1.37 x 10¹⁶ g
Φ = 166.54 degree
ρ = 3 cm⁻³
w = 500 km/s

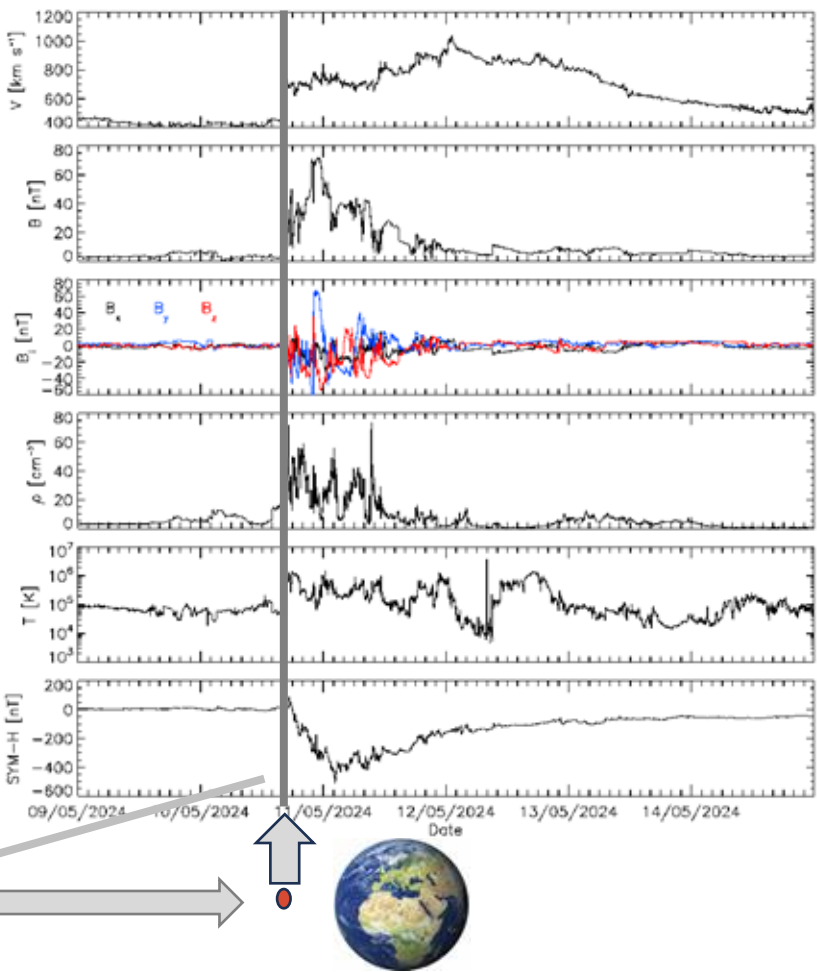
LASCO start time
2024-05-08 T19:12



Predicted travel time = 44.21 ± 3.20 h
Predicted arrival time: 2024-05-10 T18:13

Prediction error of 1 minute

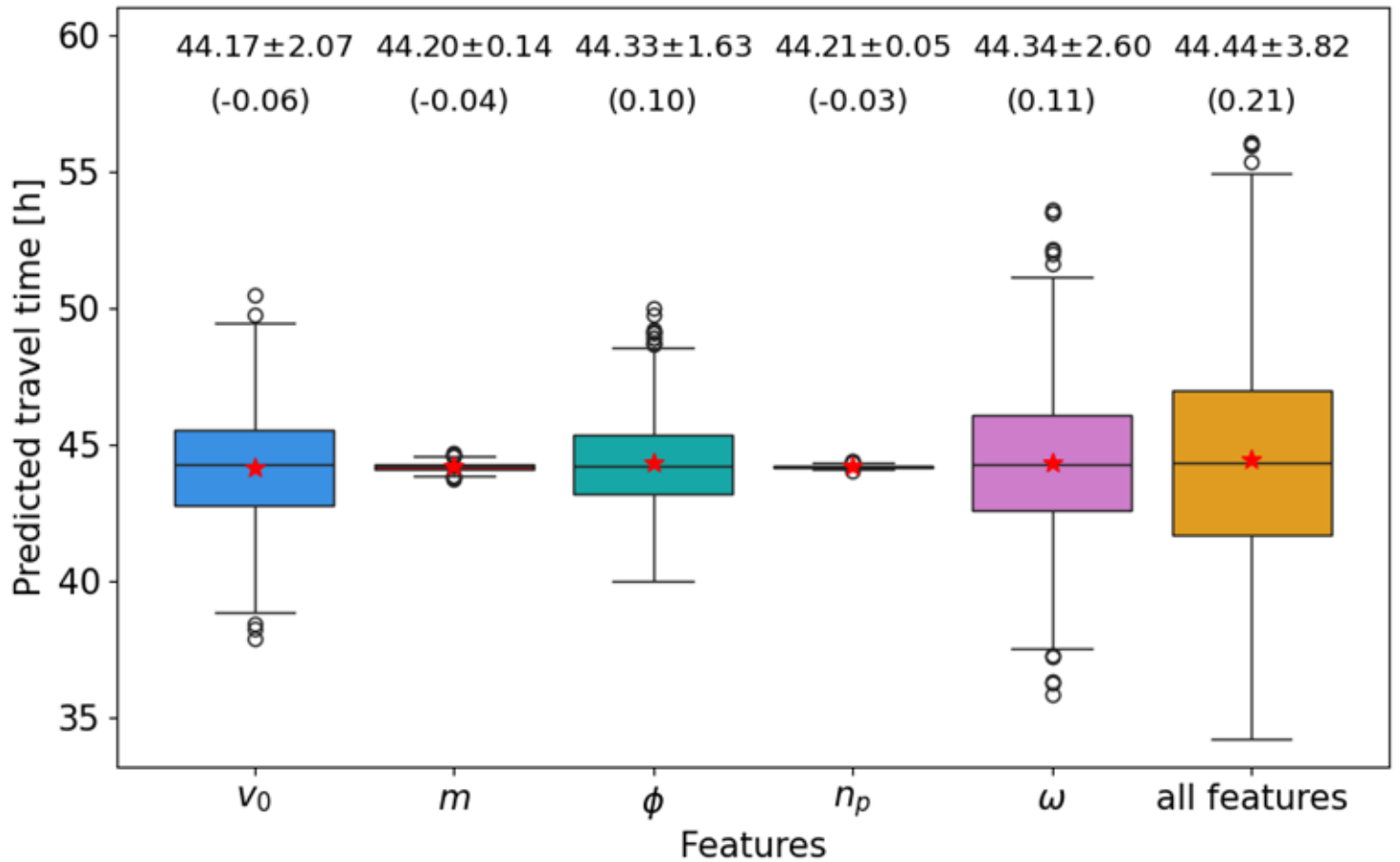
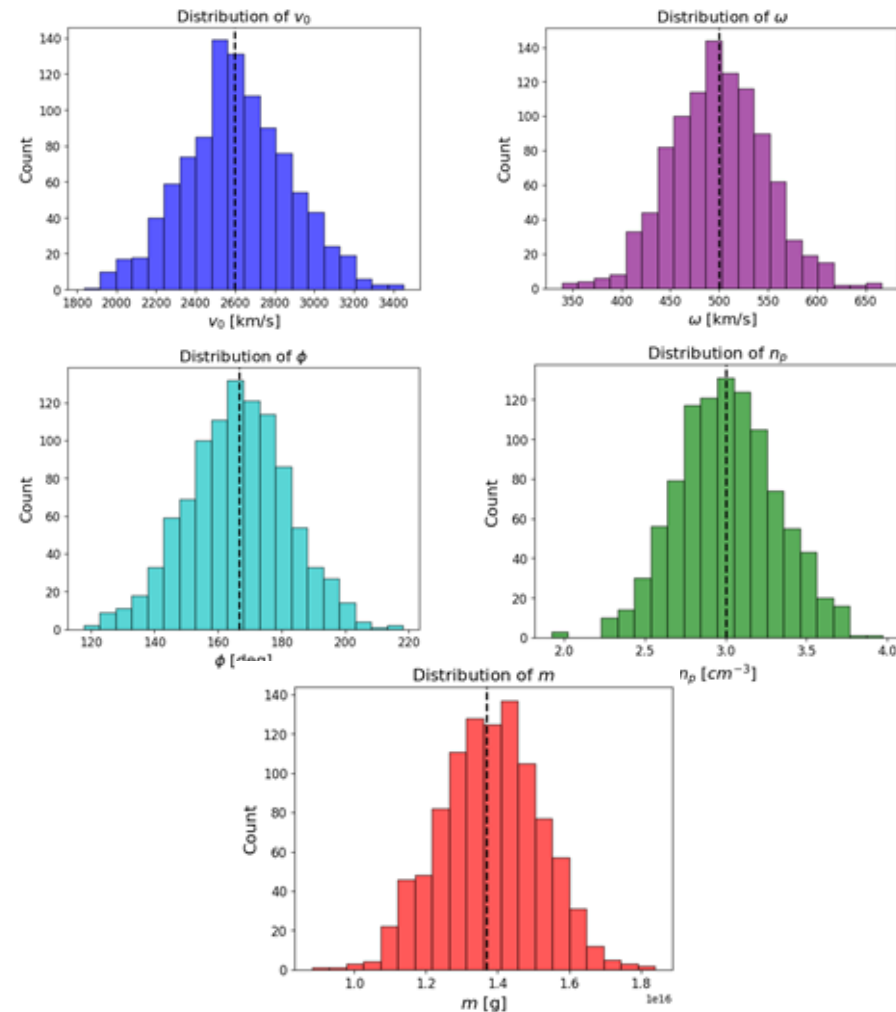
Actual arrival time = 2024-05-10 T18:14
Actual travel time = 44.23 h



L1 point

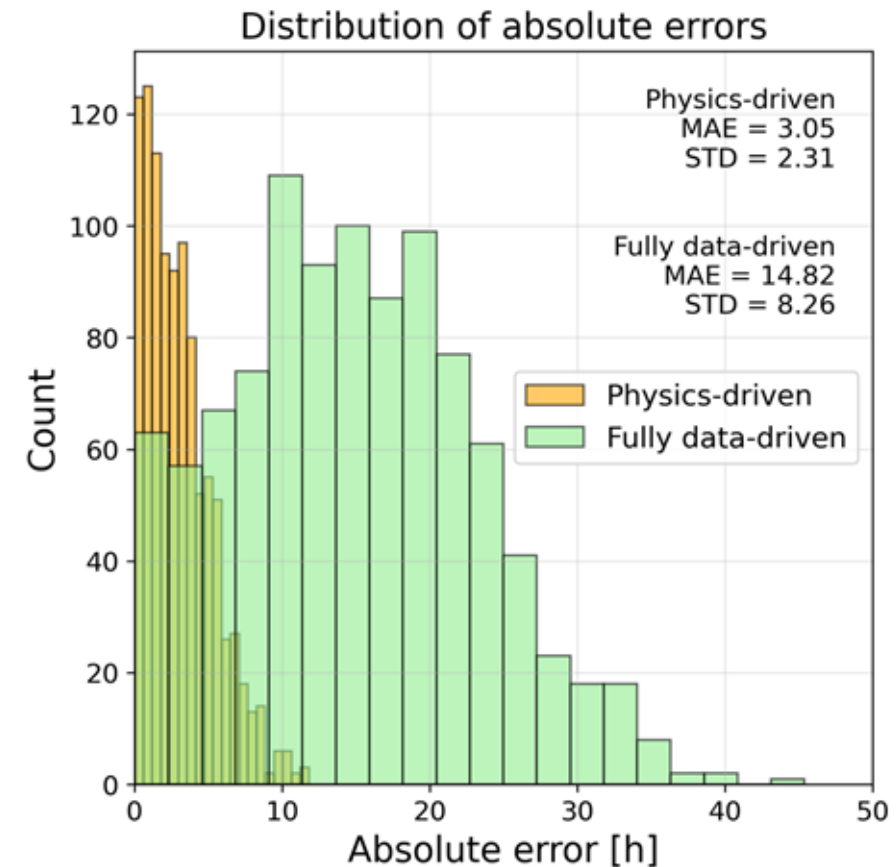
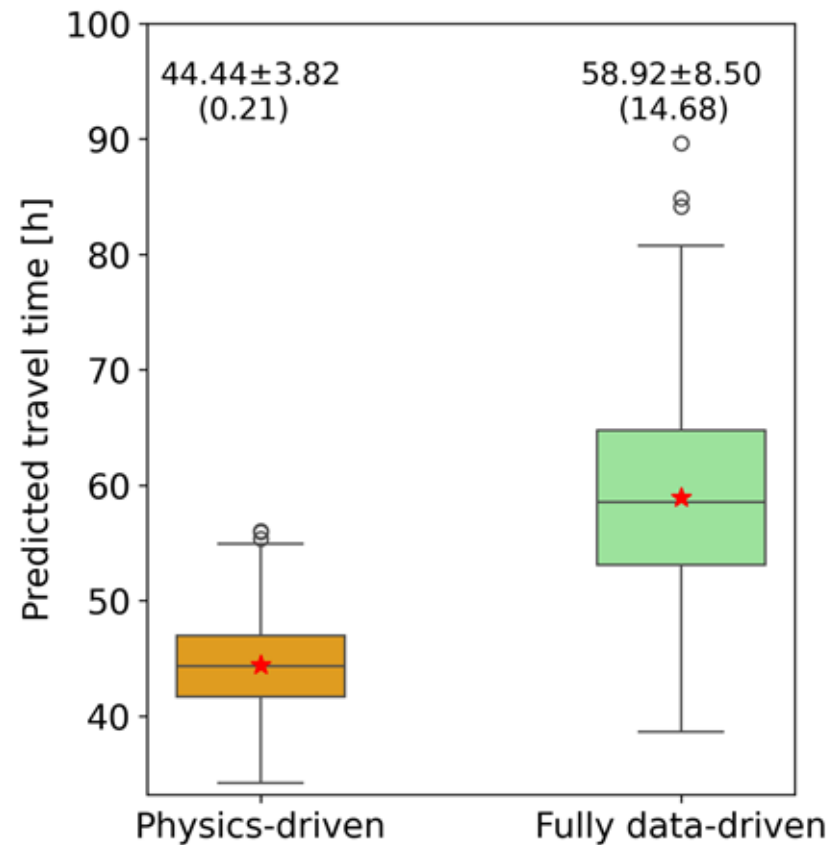
How robust is the prediction?

Actual travel time = 44.23 h



Comparison with fully data-driven method

	Physics-driven ML	Fully data-driven
Pred travel time	44.21 ± 3.20	58.64 ± 4.28
	-0.02	14.41



Moral: Physics-driven ML can improve fully data driven approaches and provides good CME arrival predictions, making it well-suited for operational space weather forecasting.

Active region classification:

- Legnaro, E., Guastavino, S., Piana, M., Massone, A.M., *Deep Learning for Active Region Classification: A Systematic Study from Convolutional Neural Networks to Vision Transformers*, ApJ **981** 157 (2025)
- Legnaro, E., Wright, P. J., Murray, S. A., Guastavino, S., Piana, M., Massone, A. M., & Maloney, S. A. (2026). ARCAFF: Cutout classification dataset. *Astronomy & Astrophysics*, 710, A91.

Flare forecasting with video-based deep learning

- Guastavino S., Marchetti F., Benvenuto F., Campi C., Piana M., *Implementation paradigm for supervised flare forecasting studies: a deep learning application with video data*, A&A, vol. 662, iss. A105, 2022
- Guastavino S., Marchetti F., Benvenuto F., Campi C., Piana M., *Operational solar flare forecasting via video-based deep learning*, Frontiers in Astronomy and Space Sciences, Vol 9, 2023

Geomagnetic storm prediction via machine learning

- Guastavino S., Bahamazava, K., Perracchione, E., Camattari, F., Audone, G., Telloni, D., Susino, R., Nicolini, G., Fineschi, S., Piana, M., Massone, A.M., *Forecasting Geo-effective Events from Solar Wind Data and Evaluating the Most Predictive Features through Machine Learning Approaches*, ApJ 971 94 (2024)
- Camattari, F., Guastavino S., Telloni, D., Perracchione, E., Piana, M., Massone, A.M., *Multi-Horizon Forecasting of Geomagnetic Disturbances: A Machine Learning Approach with Feature Ranking*, (submitted)

May 2024 Superstorm

- Guastavino S., Legnaro E., Massone A.M., Piana M., *Artificial Intelligence for the Characterization of the May 2024 Superstorm: Active Region Classification, Flare Forecasting, and Geomagnetic Storm Prediction*, ApJ (2025)
- Guastavino S., Legnaro E., Massone A.M., Piana M., *Estimating Coronal Mass Ejection Arrival with Ensemble Physics-Driven Machine Learning: The May 2024 Superstorm Case*, Solar Physics, 301(4), 53.

Physics can inform neural networks not only through the loss function, but also through the architecture itself.

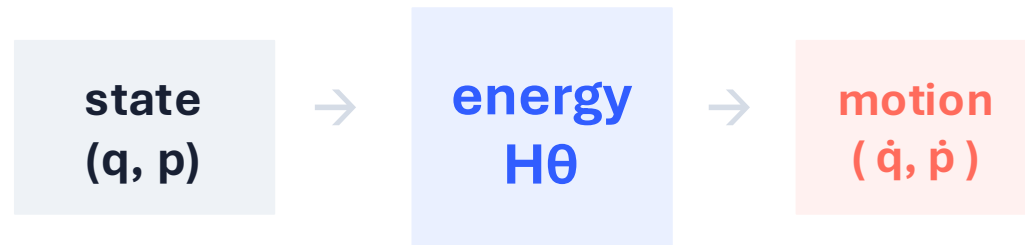
PINNs impose equations during training;

Physics-inspired architectures build physical structure directly into the model.

$$H_{\theta}(q, p)$$

$$\dot{q} = \partial H_{\theta} / \partial p \quad \dot{p} = -\partial H_{\theta} / \partial q$$

Energy-Conserving Trajectories
($\dot{q}, \dot{p}$)



Physics shapes what the model is allowed to represent.
Architecture as inductive bias

Hamiltonian Neural Networks

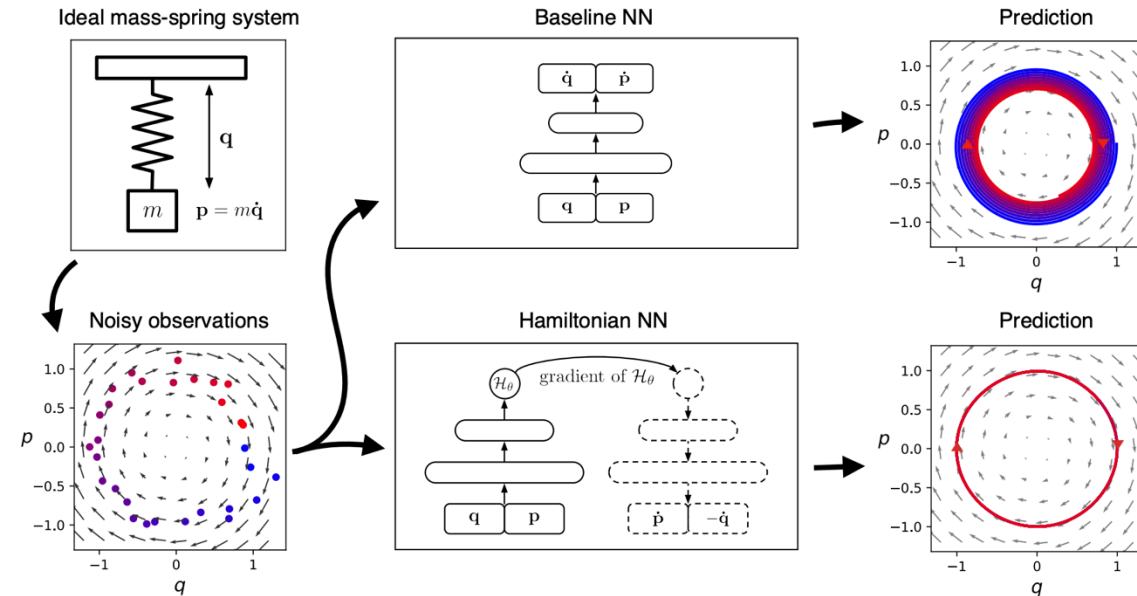
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Abstract

Even though neural networks enjoy widespread use, they still struggle to learn the basic laws of physics. How might we endow them with better inductive biases? In this paper, we draw inspiration from Hamiltonian mechanics to train models that learn and respect exact conservation laws in an unsupervised manner. We evaluate our models on problems where conservation of energy is important, including the two-body problem and pixel observations of a pendulum. Our model trains faster and generalizes better than a regular neural network. An interesting side effect is that our model is perfectly reversible in time.



Where are HNNs used?

Hamiltonian Neural Networks are used when we want to learn the dynamics of a physical system from data, but still keep the model consistent with energy-based mechanics.

They are especially useful in systems where long-term prediction matters and a generic neural network may slowly create unphysical energy drift.

Robotics and control

Hamiltonian-based Neural ODEs have been used to learn rigid-body dynamics on $SE(3)$, for systems such as pendulums, rigid bodies, and quadrotors. The advantage is that the learned model respects geometric constraints and conserves total energy by construction, unlike a black-box Neural ODE.

Molecular machine learning

HamNet uses Hamiltonian-inspired position–momentum dynamics to preserve 3D molecular conformations and build molecular fingerprints.



Thanks!

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